

7	(a)	<u>Net force acting</u> on the body and <u>the net torque acting</u> on the body is <u>zero</u> .		B1 B1
	(b)	By considering moments about the left end of the rod: $0.25L \times 400 + 0.50L \times 120 = T_2 \cos 30^\circ \times L$ $T_2 = 185 \text{ N}$ horizontally: $185 \sin 30^\circ = T_1 \sin 14.4^\circ$ $T_1 = 371 \text{ N}$ OR $T_2 \cos 30^\circ + T_1 \cos 14.4^\circ = 400 + 120$ $T_1 = 371 \text{ N}$ OR Considering moments about right end of rod: $0.75L \times 400 + 0.50L \times 120 = T_1 \cos 14.4^\circ \times L$ $T_1 = 372 \text{ N}$		C1 A1 C1 A1
	(c)	The frequency f of a body undergoing simple harmonic motion refers to the number of oscillation it undergoes per unit time and is measured in cycles per second. The angular frequency ω is the angular velocity of the body in SHM measured in radians per second and $\omega = 2\pi f$.		B1 B1
	(d)	(i)	When the spring is displaced by x , one of the springs (left) is compressed and the other spring (right) is extended. Both springs will exert a force of $-2kx$ on the mass. $F = ma = -2kx$ $a = -\frac{2kx}{m}$	B1 B1

		(ii)	$E_T = \frac{1}{2} m \omega^2 x_0^2 = \frac{1}{2} \times 0.038 \times (2\pi \times 3.2)^2 \times 0.028^2$ $= 0.00602 \text{ J} = 6.0 \text{ mJ (2 sf)}$	M1 A1
		(iii)	When K.E. of ball is equal to P.E. of spring, $E_P = \frac{1}{2} E_T \quad \text{OR} \quad E_P = E_K$ $\frac{1}{2} m \omega^2 x^2 = \frac{1}{2} E_T \quad \frac{1}{2} m \omega^2 x^2 = \frac{1}{2} m \omega^2 (x_0^2 - x^2)$ $x = \sqrt{\frac{E_T}{m \omega^2}} \quad x = \frac{x_0}{\sqrt{2}}$ $= \sqrt{\frac{0.00602}{0.038 \times (2\pi \times 3.2)^2}} \quad = \frac{2.8}{\sqrt{2}}$ $= 0.0198 \text{ m} = 2.0 \text{ cm}$ $= 1.98 \text{ cm} = 2.0 \text{ cm} = 0.020 \text{ m}$ Note: Distance can only be positive.	C1 C1 A1
		(iv)	Fig. 2.2 T: Horizontal line at 6 mJ K: max curve between 0 to 6 mJ	B1 B1 B1

			P: min curve between 0 to 6 mJ Deduct 1m for (max 3m): Any missing labels K & P not intersecting at 3 mJ (not marking for 2.0 cm) Any graphs not starting and ending at ± 2.8 cm.	
	(e)	In a vertical spring mass system, there is a change in gravitational potential energy. At the highest position, EPE will be reduced. or at the equilibrium position, the KE will be reduced.	B1 B1	